Greater Glider Monitoring Plan Design

- How many Sites?
- Where to put them?



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How many sites?

- Power Analysis is used to determine whether the Monitoring Plan has the capacity to detect change, if it is occurring?

Considerations:

- How much change is unacceptable (e.g. 10%, 30%, or 50% decline)
  - Effect size is the measure of difference, or change, that is important to identify
- How much confidence do you need that the change is actually occurring, as opposed to chance
  - Power is the probability that a monitoring plan can distinguish an actual effect from a chance occurrence
- What is the likely variability of glider counts at sites within the region of interest
- # Sites required is directly related to the above considerations:
  - E.g. very large numbers of monitoring sites are required to detect small changes, with high confidence (statistical power), in highly variable environments – financial constraints may apply

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Power analysis

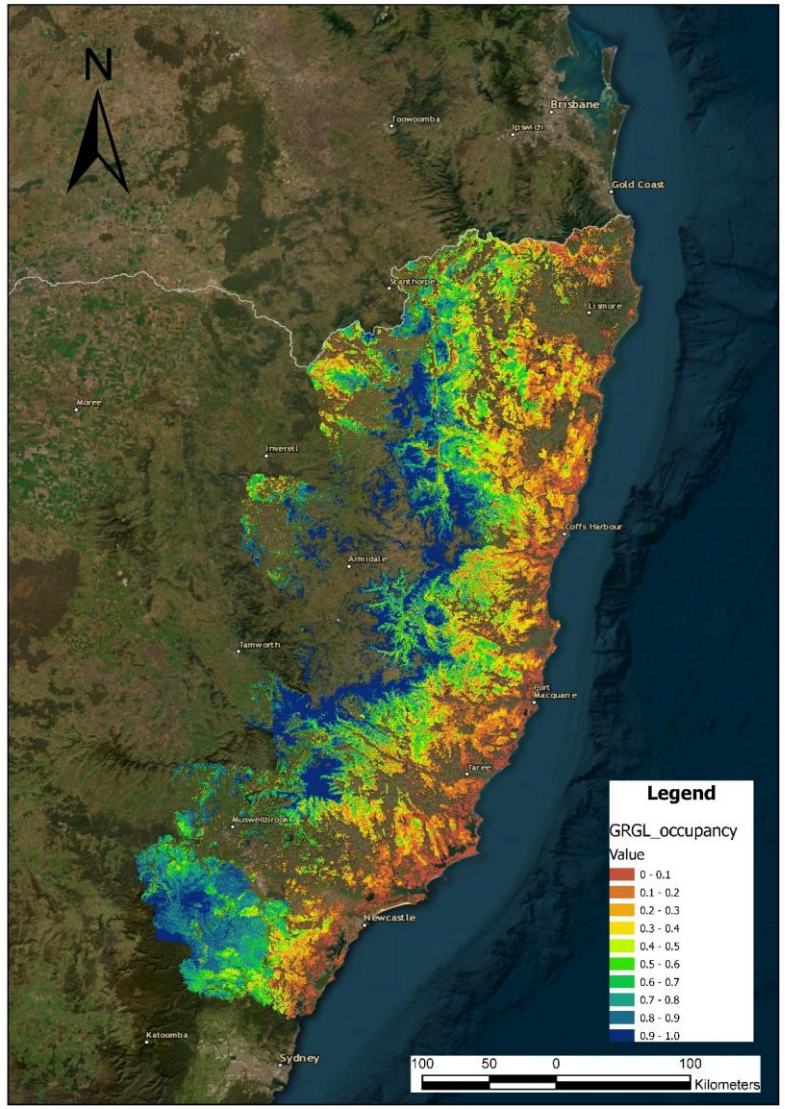
A priori power analysis for two-tailed t-tests assessing the difference between two independent means showing the implications for sample size requirements of variations in desired power and effect size.

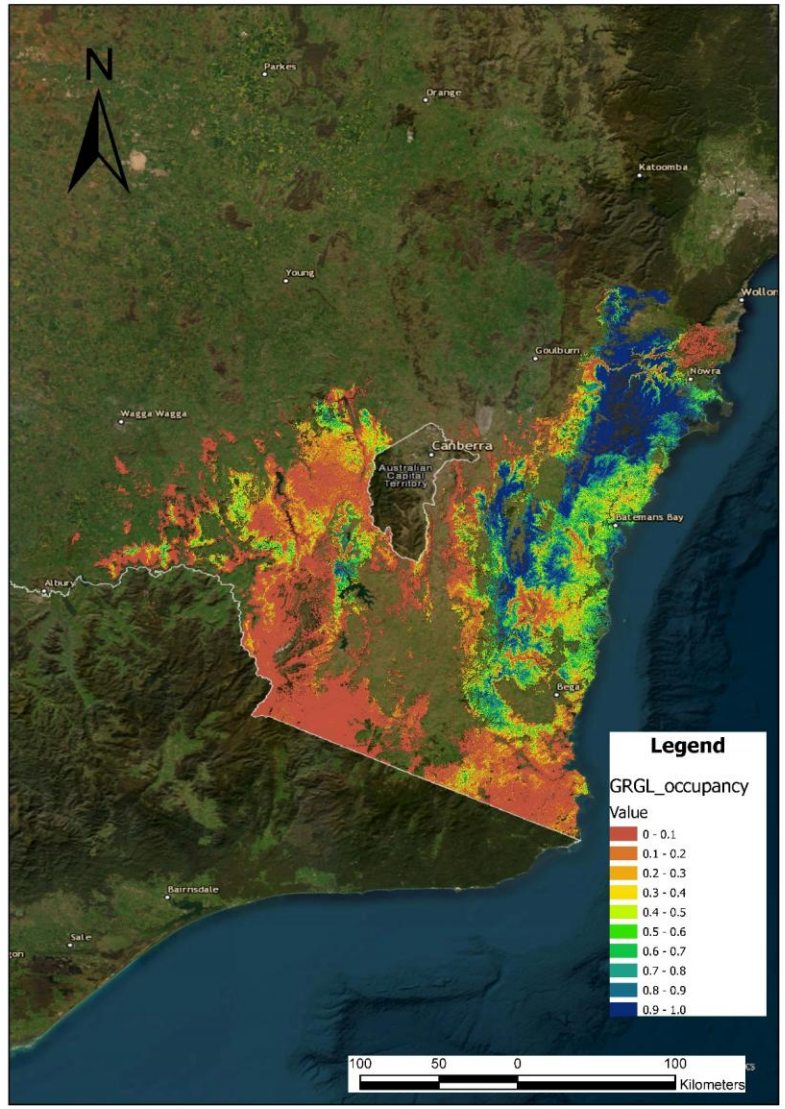
[1]Note: The magnitude of change detectable, and its correlation with Cohen's d-values, was estimated using population mean and standard deviation data recorded from Greater Glider post-fire surveys in 2020-2021 in north-eastern NSW (Kavanagh *et al*. 2026).

Power	Alpha	Effect size (Cohen's d)	Sample size (no. sites)	[1]Approx. change detectable (%)
0.80	0.05	0.2	394	18.8
0.85	0.05	0.2	450	18.8
0.90	0.05	0.2	527	18.8
0.95	0.05	0.2	651	18.8
0.80	0.05	0.3	176	28.8
0.85	0.05	0.3	201	28.8
0.90	0.05	0.3	235	28.8
0.95	0.05	0.3	290	28.8
0.80	0.05	0.4	100	37.5
0.85	0.05	0.4	114	37.5
0.90	0.05	0.4	133	37.5
0.95	0.05	0.4	164	37.5

Variability in Greater Glider Occupancy

Statistical Power is improved if monitoring sites are located in generally good habitat (e.g. where probability of Greater Glider occurrence > 0.75).
i.e. Fewer sites are required to detect changes if they are occurring.





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Where to put Greater Glider monitoring sites?

Stratify site locations by:

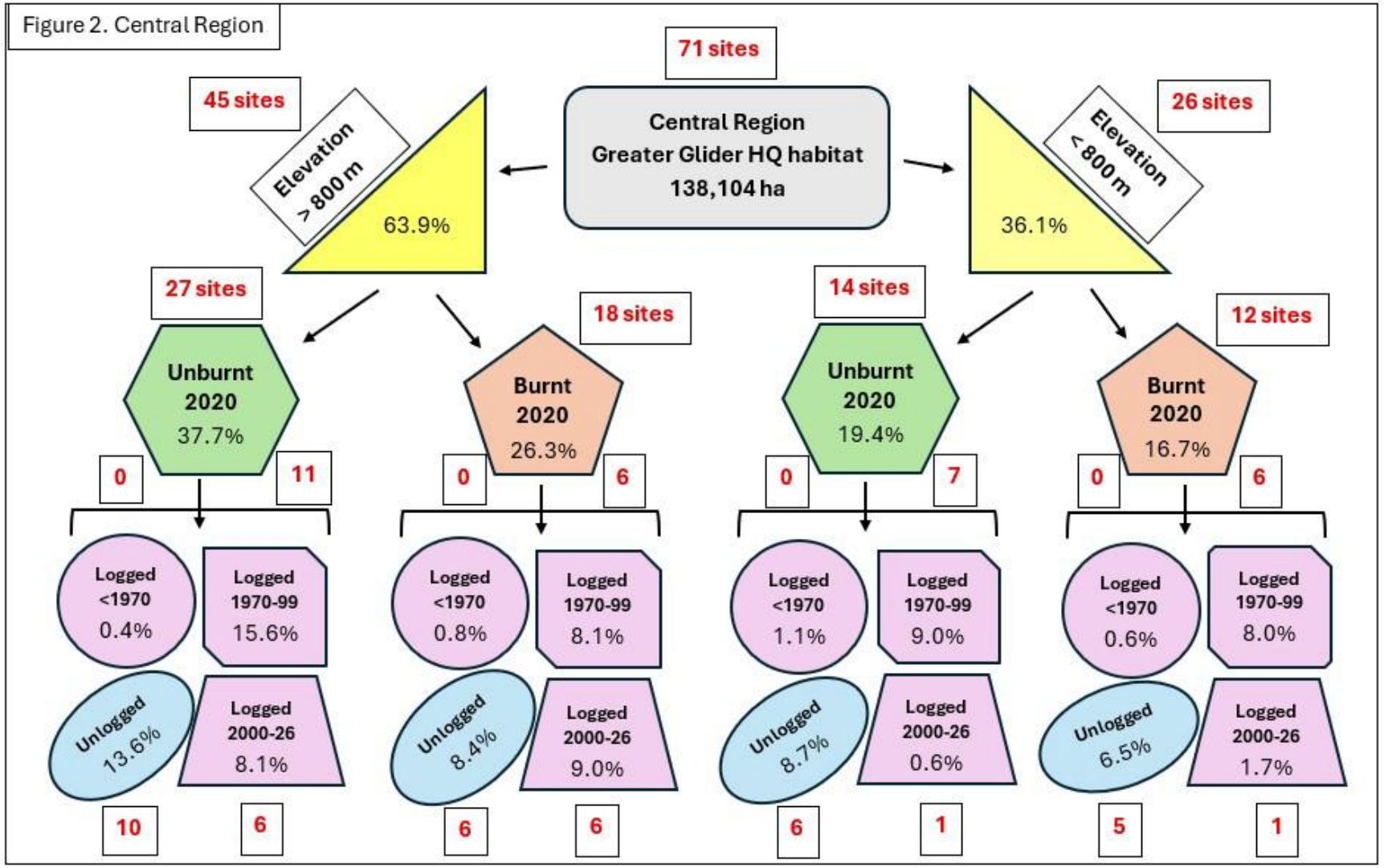
- GG probability of occurrence (P>0.75) using the 2022 Occupancy Model built using the comprehensive early-mid 1990s (pre-climate change) datasets
- Elevation (surrogate for temperature and rainfall) – major driver of GG habitat quality
- Fire history - Unburnt and burnt sites affected by the severe 2019-2020 wildfires
- Logging history – Sites with no recorded history of logging and those with old and more recent history of logging

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Proportion of area (percent) and number of sites allocated per stratification category in Central Region

(Total sites across 3 regions=176 sites)

Note: the category ‘last logged <1970’ is very uncommon and is not used.



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Summary of monitoring-site selection results.

Number of sites required in each stratum across three CIFOA regions in NSW. (Data based on proportional availability of each stratum).

Stratum/Sites	Southern Region	Central Region	North-East Region	Total
< 800, 1970-1999, Burnt	4	6	7	17
< 800, 1970-1999, Unburnt	3	7	5	15
< 800, 2000-2026, Burnt	2	1	5	8
< 800, 2000-2026, Unburnt	1	1	2	4
< 800, No harvest history, Burnt	10	5	3	18
< 800, No harvest history, Unburnt	8	6	5	19
> 800, 1970-1999, Burnt	2	6	6	14
> 800, 1970-1999, Unburnt	2	11	6	19
> 800, 2000-2026, Burnt	2	6	7	15
> 800, 2000-2026, Unburnt	2	6	5	13
> 800, No harvest history, Burnt	3	6	4	13
> 800, No harvest history, Unburnt	6	10	5	21
Total	45	71	60	176

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Distribution of 209 Greater Glider trend monitoring sites in Coastal IFOA regions of NSW

(includes 33 ‘spare’ sites)

